Research Project - Mathematical Statistics

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1 Introduction

The admission process for graduate studies is a critical step in defining our future careers, as a consequence, it is filled with ambition as the volume of candidates competes for limited spots in prestigious programs. Among the factors that admission committees have to analyze, there are both quantitative (GRE, TOEFL, GPA, etc.) and qualitative metrics (recommendation letters, personal statements, etc.). As students deeply involved in this process, we were driven by curiosity to discover what sets successful candidates apart.

Our study takes a systematic approach, delving into the aspects of admissions and carefully analyzing the contributions of each factor. This project, therefore, holds the potential to provide key insights that may guide future applicants in optimizing their profiles for success. The competitive landscape of higher education, marked by an ever-growing pool of talented individuals, makes this endeavor both timely and significant.

2 Dataset

The dataset used in this analysis, titled *Graduate Admission 2*, was sourced from Kaggle and is distributed under the CC0 public license, meaning it is in the public domain. For transparency and reproducibility, the link to the dataset will be included in the bibliography section of this report (Bibliography).

This dataset is inspired by the UCLA Graduate Dataset, widely known in academic circles for admissions-related studies. It is important to note that the test scores and GPA in this dataset are presented in their older formats, which differ from the formats currently in use.

The dataset consists of 500 observations, each corresponding to an individual applicant to a graduate program. The observations provide a structured representation of applicants' profiles, including quantitative metrics and a target variable indicating the likelihood of admission. Specifically, the dataset includes the following variables:

1. **Serial No.**: Identification number of application.
2. **GRE Score**: Applicant's Graduate Record Examination score [0, 340].
3. **TOEFL Score**: Applicant's Test of English as a Foreign Language score [0, 120].
4. **University Rating**: A ranking of the university from which the applicant graduated [1, 5].
5. **SOP**: Applicant's Statement of Purpose [1, 5].
6. **LOR**: Applicant's Letters of Recommendation [1, 5].
7. **CGPA**: Applicant's undergraduate Grade Point Average [0, 10].
8. **Research**: A binary variable indicating whether the applicant has prior research experience (1 for yes, 0 for no).
9. **Chance of Admission**: A continuous variable ranging from 0 to 1, representing the predicted probability of admission to the program.

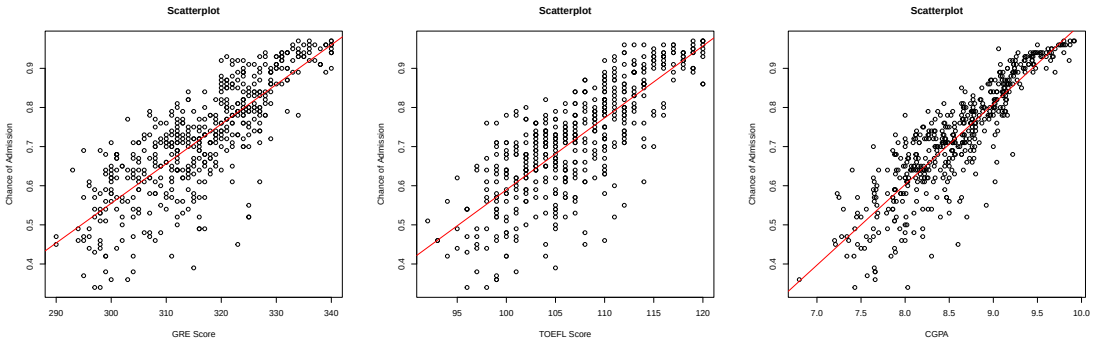
2.1 Data Cleaning

Before moving on to data analysis, we performed what is known as *Data Cleaning*. These steps were essential to ensure that the data was ready for exploration, reducing the risk of encountering errors later on. The first column, serial number, was removed as it was not useful for our study; it served only as an identifier and did not provide any predictive value. The dataset was then checked for any missing values, and since none were found, we were able to proceed.

Boxplots of the continuous variables were used to detect outliers, as these kinds of extreme values can skew results and impact the accuracy of the model. Fortunately, no significant outliers were identified. Next, we reviewed the data for its format: histograms of continuous variables showed that they were on meaningful scales, and the binary variable, Research, was encoded using only 0/1 as unique values, making it suitable for our study without further adjustments. These steps ensured that our dataset was clean, consistent, and ready for statistical modeling. (Appendix A.1)

2.2 Data Visualization

Data visualization plays a vital role in understanding relationships and distributions within the dataset, offering valuable insights that guide further analysis. To explore the connections between the target variable, Chance of Admit, and each predictor, scatterplots were created for all combinations (Appendix A.1), the following are the most significant ones.



3 Regression Model

The objective of this section is to estimate the relationships between the predictors and the target variable *Chance of Admit* using a multiple linear regression model. This approach allows us to quantify the impact of each predictor on the likelihood of admission while accounting for the effects of other variables. A multiple linear regression model was fitted to the data using the equation:

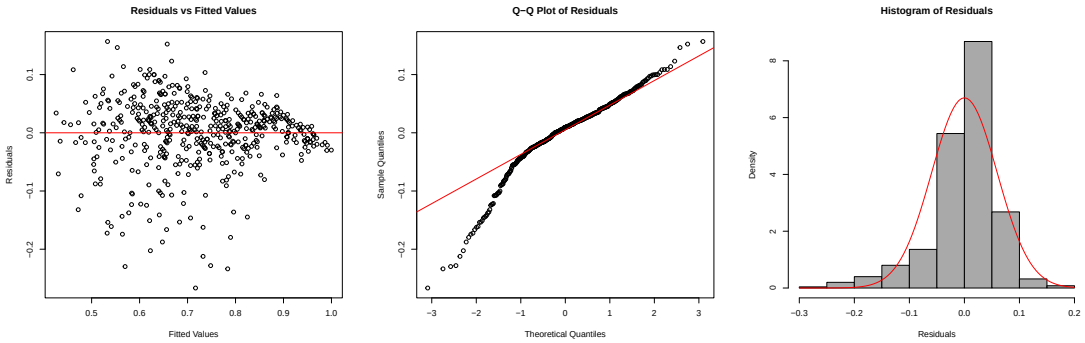
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_p X_{pi} + e_i \quad \text{with } i = 1, ..., n \quad n = 500$$

where Y_i represents the *Chance of Admit* for the i-th observation, $X_{1i}, X_{2i}, \ldots, X_{pi}$ are the predictors, β_j are the regression coefficients representing the effect of each predictor, and ϵ_i is the random error term for the i-th observation.

3.1 Validity assumptions

Before interpreting the results, we run diagnostic checks to validate the model's assumptions and ensure its reliability. The QQ plot of the residuals indicates that they are approximately normally distributed, even though there are minor deviations at the tails. This suggests that normality is mostly but not entirely satisfied, as showed by the Shapiro-Wilk test. The histogram of residuals further supports this observation, showing a roughly symmetric bell-shaped distribution centered around zero.

The residuals versus fitted values plot shows no clear pattern, which suggests that the assumption of homoscedasticity is satisfied, as the residual variance appears constant across all fitted values. However, there is some slight non-linearity observable in the residuals, indicating that the assumption of linearity between predictors and the target variable is only partially met. In addition, multicollinearity was examined among predictors to ensure that no predictor was excessively correlated with another. (Appendix A.2).

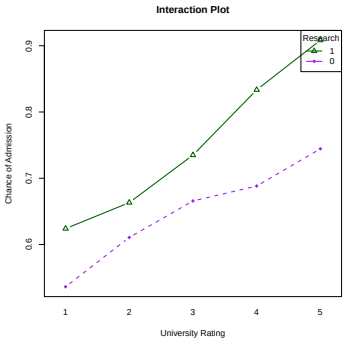


3.2 Interaction Effects

Interaction effects are crucial in regression analysis as they capture interdependencies between predictors and the response variable. To explore the potential interaction between Research and University Rating, an interaction plot was first created to visualize their combined effect on Chance of Admit. The plot suggested a dependency, prompting us to include an interaction term in the model for further analysis.

An **ANOVA** test comparing the baseline model and the interaction model confirmed a significant improvement ($F = 5.1605$, $p = 0.02354$), rejecting the null hypothesis that the interaction term does not enhance the model. While the improvement in adjusted R^2 was marginal, increasing from 0.8194 to 0.8209, and the reduction in residual standard error was minimal, the interaction term added interpretive depth. The coefficient for the interaction term ($\beta = 0.0123$, $p = 0.02354$) indicates that the relationship between University Rating and Chance of Admit is influenced by research experience. (Appendix A.2)

Although the improvement in model performance was slight, the interaction term provides valuable insights into the interplay between predictors. Applicants with prior research experience from top-rated universities may have an advantage, as admissions committees often correlate such profiles with stronger academic and professional preparation.




```
Analysis of Variance Table

Model 1: data$Chance.of.Admit ~ data$GRE.Score + data$TOEFL.Score + data$University.Rating +
    data$SOP + data$LOR + data$CGPA + data$Research
Model 2: data$Chance.of.Admit ~ data$Research * data$University.Rating +
    data$CGPA + data$TOEFL.Score + data$SOP + data$LOR + data$GRE.Score +
    data$Research
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1    492 1.7704
2    491 1.7520  1  0.018413 5.1605 0.02354 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

4 Model Selection

4.1 Step-down method

To improve the regression model and ensure it balances simplicity and predictive accuracy, we decided to apply a step-down method based on backward elimination. Starting with the full model, which includes all main effects and the significant interaction term, this approach will iteratively tests the null hypothesis $H_0 : \theta_i = 0$ for each parameter. If the *p-value* for a parameter exceeds the chosen significance threshold (0.05), the parameter is removed, and the model is refitted with the remaining parameters. This process continues, eliminating one parameter at a time, until all remaining parameters are statistically significant.

The parameter SOP was removed due to its lack of statistical significance (p = 0.64) in the full model. In order to evaluate model performance we conducted a comparison between the full and the refined model. The full model had an adjusted R^2 of 0.8209, while the refined model achieved a slightly improved adjusted R^2 of 0.8212, indicating that the removal of non-significant terms did not compromise explanatory power. In terms of model fit criteria, the refined model outperformed the full model with a lower **AIC** and **BIC**, confirming that the refined model achieves a better balance between fit and simplicity. The residual standard error for the refined model was 0.05969, compared to 0.05973 for the full model, indicating a marginal improvement in residual variance. We then run a diagnostic over the refined model to check the validity of the assumptions. (Appendix A.3)

	df	AIC
interaction_model	10	-1387.997
step_model	9	-1389.771

	df	BIC
interaction_model	10	-1345.851
step_model	9	-1351.840

4.2 Cross Validation

The objective of this analysis was to validate the refined regression model and assess its predictive power using k-fold cross-validation. We selected 10-fold cross-validation as a balance between bias and variance in model evaluation. Using fewer folds, such as 5, might introduce higher variance in the results because each test set would contain fewer observations. Conversely, increasing the number of folds (e.g., 20-fold) could reduce variance but significantly increase computational cost without substantial gains in model performance. The 10-fold approach offers an optimal trade-off between computational

efficiency and the robustness of results. The model was trained on 9 folds and tested on the remaining one, with this process repeated across all folds, ensuring that every observation is used for both training and testing. The results showed an average R^2 of 0.820, indicating the model explains a significant portion of the variance in Chance of Admit, with an RMSE of 0.060. The low standard deviations for R^2 and RMSE confirm the model's predictive power.

```
  intercept       RMSE  Rsquared        MAE      RMSESD RsquaredSD       MAESD
1      TRUE 0.06008634 0.8200829 0.04330882 0.005539359 0.05100248 0.002949964
```

5 Limitations and conclusions

This project provided insights into which factors mostly influence the chance of admission to graduate programs, while also validating a predictive model. From this analysis, it emerged that the most significant predictors are academic credentials, such as GPA, GRE and TOEFL, suggesting that candidates should prioritize excelling academically. Nevertheless research experience, combined with university rating, remains significant. The final model, validated through 10-fold cross-validation, exhibited strong predictive power, explaining over 82% of the variance in chance of admission with minimal prediction errors, confirming its reliability and generalizability.

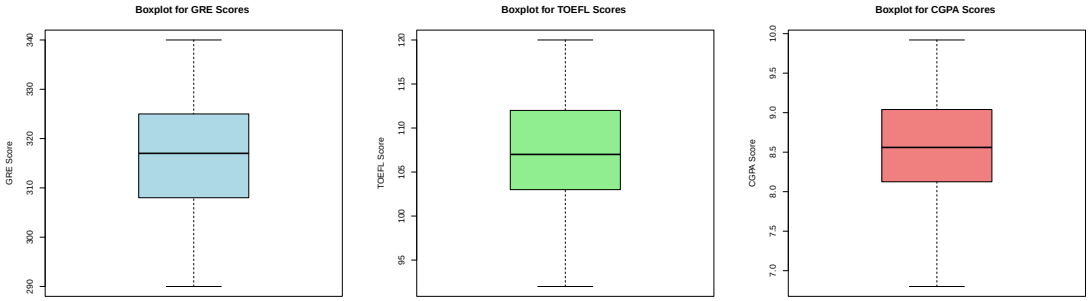
The analysis encountered several limitations that may have influenced the results. Although the dataset contained 500 observations and captured academic and personal variables, it may still be too small to capture more complex relationships. Moreover, the multicollinearity between variables, such as GRE and TOEFL scores, may have introduced bias in the estimates of the coefficients. The assumptions of linear regression were partially met; although the residual plots showed approximate linearity and homoscedasticity, the Shapiro-Wilk test revealed a deviation from normality that may cause issues with the reliability of confidence intervals and hypothesis tests, particularly in small datasets..

These results underline the importance of academic performance in shaping admissions outcomes, while also highlighting opportunities for further research. In order to improve the accuracy of this predictive model, one could adopt different strategies such as non-parametric bootstrapping to generate robust confidence intervals and p-values, data transformations (e.g., log) or non-linear models to better capture complex relationships and address residual non-normality. Additionally, investigating the practical significance of interaction effects and employing more diverse validation techniques would provide a deeper understanding of the factors driving admission outcomes. This study serves as a rigorous foundation for future analyses, combining theoretical and empirical rigor to inform the graduate admissions process.

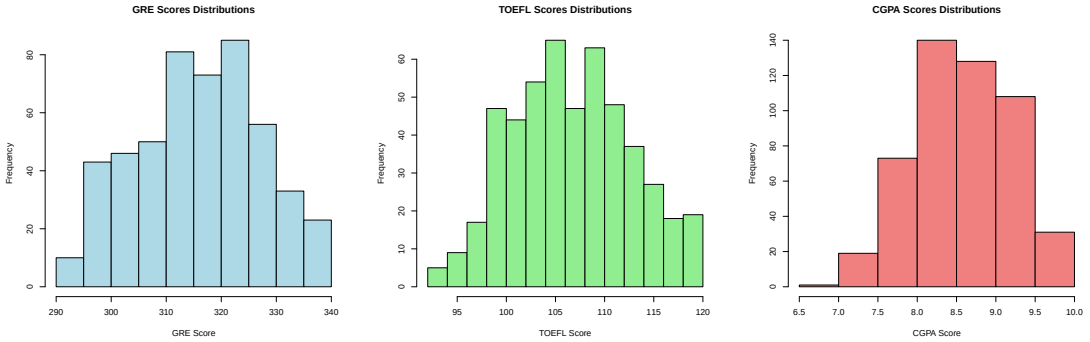
A Appendix : Supplementary Material

A.1 Data Set

- Boxplots for checking outliers



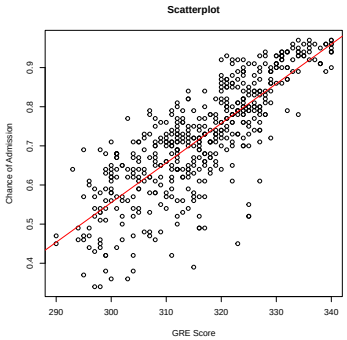
- Histograms for continuous variables scale and distribution

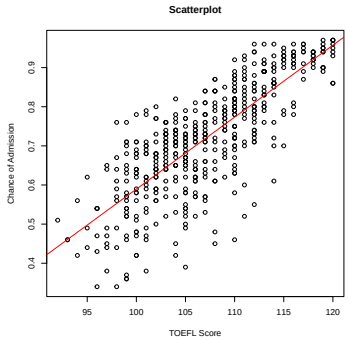


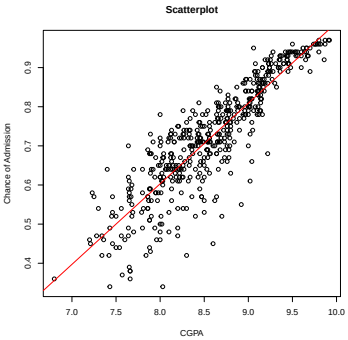
- First summary of the data

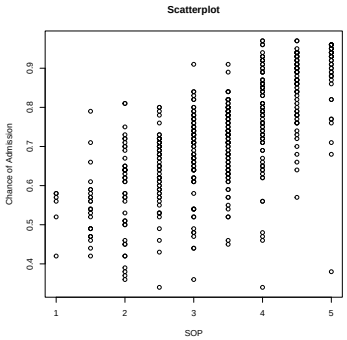
```
   GRE.Score      TOEFL.Score    University.Rating      SOP
 Min.   :290.0   Min.   : 92.0   Min.   :1.000     Min.   :1.000
 1st Qu.:308.0   1st Qu.:103.0   1st Qu.:2.000     1st Qu.:2.500
 Median :317.0   Median :107.0   Median :3.000     Median :3.500
 Mean   :316.5   Mean   :107.2   Mean   :3.114     Mean   :3.374
 3rd Qu.:325.0   3rd Qu.:112.0   3rd Qu.:4.000     3rd Qu.:4.000
 Max.   :340.0   Max.   :120.0   Max.   :5.000     Max.   :5.000
      LOR             CGPA           Research    Chance.of.Admit
 Min.   :1.000   Min.   :6.800   Min.   :0.00   Min.   :0.3400
 1st Qu.:3.000   1st Qu.:8.127   1st Qu.:0.00   1st Qu.:0.6300
 Median :3.500   Median :8.560   Median :1.00   Median :0.7200
 Mean   :3.484   Mean   :8.576   Mean   :0.56   Mean   :0.7217
 3rd Qu.:4.000   3rd Qu.:9.040   3rd Qu.:1.00   3rd Qu.:0.8200
 Max.   :5.000   Max.   :9.920   Max.   :1.00   Max.   :0.9700
```

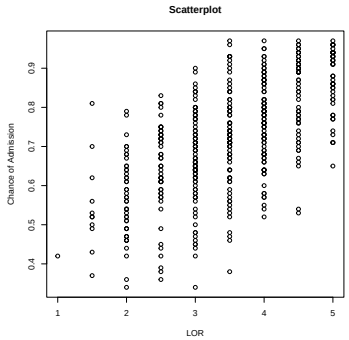
- Scatter plots

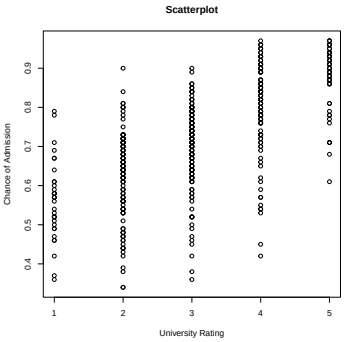


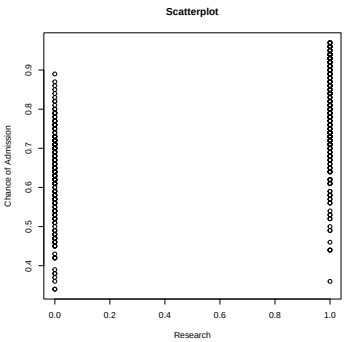












A.2 Regression Model

- Summary of the Multiple Linear Regression Model

```
Call:
lm(formula = data$Chance.of.Admit ~ data$GRE.Score + data$TOEFL.Score +
    data$University.Rating + data$SOP + data$LOR + data$CGPA +
    data$Research, data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.266657 -0.023327  0.009191  0.033714  0.156818

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)           -1.2757251  0.1042962 -12.232  < 2e-16 ***
data$GRE.Score         0.0018585  0.0005023   3.700 0.000240 ***
data$TOEFL.Score       0.0027780  0.0008724   3.184 0.001544 **
data$University.Rating 0.0059414  0.0038019   1.563 0.118753
data$SOP               0.0015861  0.0045627   0.348 0.728263
data$LOR               0.0168587  0.0041379   4.074 5.38e-05 ***
data$CGPA              0.1183851  0.0097051  12.198  < 2e-16 ***
data$Research          0.0243075  0.0066057   3.680 0.000259 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.05999 on 492 degrees of freedom
Multiple R-squared:  0.8219,    Adjusted R-squared:  0.8194
F-statistic: 324.4 on 7 and 492 DF,  p-value: < 2.2e-16
```

- Shapiro-Wilk test for the validity assumptions

```
        Shapiro-Wilk normality test

data:  res
W = 0.92549, p-value = 4.824e-15
```

- Correlation Matrix

```
                  GRE.Score TOEFL.Score University.Rating       SOP       LOR
GRE.Score         1.0000000   0.8272004         0.6353762 0.6134977 0.5246794
TOEFL.Score       0.8272004   1.0000000         0.6497992 0.6444104 0.5415633
University.Rating 0.6353762   0.6497992         1.0000000 0.7280236 0.6086507
SOP               0.6134977   0.6444104         0.7280236 1.0000000 0.6637069
LOR               0.5246794   0.5415633         0.6086507 0.6637069 1.0000000
CGPA              0.8258780   0.8105735         0.7052543 0.7121543 0.6374692
Research          0.5633981   0.4670121         0.4270475 0.4081158 0.3725256
Chance.of.Admit   0.8103506   0.7922276         0.6901324 0.6841365 0.6453645
                       CGPA  Research Chance.of.Admit
GRE.Score         0.8258780 0.5633981       0.8103506
TOEFL.Score       0.8105735 0.4670121       0.7922276
University.Rating 0.7052543 0.4270475       0.6901324
SOP               0.7121543 0.4081158       0.6841365
LOR               0.6374692 0.3725256       0.6453645
CGPA              1.0000000 0.5013110       0.8824126
Research          0.5013110 1.0000000       0.5458710
Chance.of.Admit   0.8824126 0.5458710       1.0000000
```

- Summary of the Interaction Model

```
Call:
lm(formula = interaction_formula, data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.266693 -0.023002  0.009158  0.033490  0.155553

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)            -1.2705887  0.1057744 -12.012  < 2e-16 ***
data$GRE.Score          0.0018508  0.0005034   3.676 0.000263 ***
data$TOEFL.Score        0.0027897  0.0008741   3.191 0.001506 **
data$University.Rating  0.0058140  0.0038287   1.519 0.129527
data$SOP                0.0017389  0.0045949   0.378 0.705267
data$LOR                0.0158150  0.0053984   2.930 0.003552 **
data$CGPA               0.1182799  0.0097204  12.168  < 2e-16 ***
data$Research           0.0178422  0.0224451   0.795 0.427039
data$LOR:data$Research  0.0019227  0.0063789   0.301 0.763220
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.06004 on 491 degrees of freedom
Multiple R-squared:  0.8219,    Adjusted R-squared:  0.819
F-statistic: 283.3 on 8 and 491 DF,  p-value: < 2.2e-16
```

A.3 Model Selection

- Step Down Method

```
Start:  AIC=-2808.94
data$Chance.of.Admit ~ data$Research * data$University.Rating +
    data$CGPA + data$TOEFL.Score + data$SOP + data$LOR + data$GRE.Score +
    data$Research

                                       Df Sum of Sq    RSS     AIC
- data$SOP                              1   0.00079 1.7528 -2810.7
<none>                                              1.7520 -2808.9
- data$Research:data$University.Rating  1   0.01841 1.7704 -2805.7
- data$TOEFL.Score                      1   0.03798 1.7899 -2800.2
- data$GRE.Score                        1   0.04475 1.7967 -2798.3
- data$LOR                              1   0.06072 1.8127 -2793.9
- data$CGPA                             1   0.51629 2.2683 -2681.8

Step:  AIC=-2810.71
data$Chance.of.Admit ~ data$Research + data$University.Rating +
    data$CGPA + data$TOEFL.Score + data$LOR + data$GRE.Score +
    data$Research:data$University.Rating

                                       Df Sum of Sq    RSS     AIC
<none>                                              1.7528 -2810.7
- data$Research:data$University.Rating  1   0.01806 1.7708 -2807.6
- data$TOEFL.Score                      1   0.03938 1.7921 -2801.6
- data$GRE.Score                        1   0.04444 1.7972 -2800.2
- data$LOR                              1   0.07116 1.8239 -2792.8
- data$CGPA                             1   0.54250 2.2953 -2677.9
```

- Summary of the Step Model

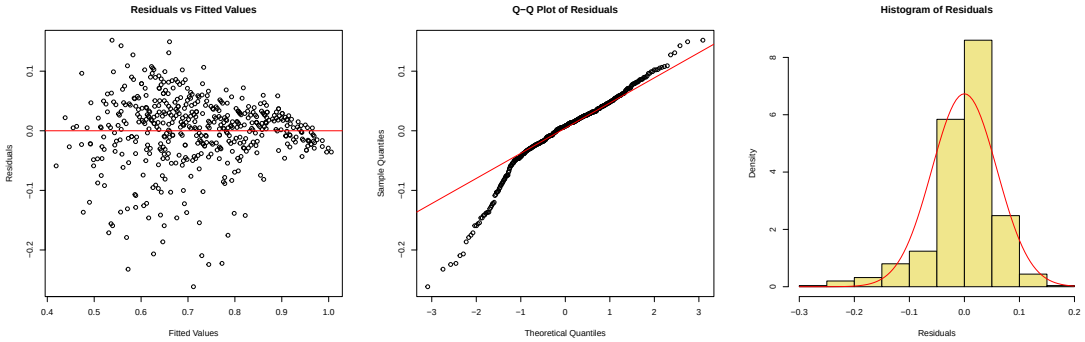
```
Call:
lm(formula = data$Chance.of.Admit ~ data$Research + data$University.Rating +
    data$CGPA + data$TOEFL.Score + data$LOR + data$GRE.Score +
    data$Research:data$University.Rating)

Residuals:
      Min        1Q    Median        3Q       Max
-0.261663 -0.024024  0.007733  0.032898  0.151950

Coefficients:
                                        Estimate Std. Error t value Pr(>|t|)
(Intercept)                           -1.2315778  0.1052693 -11.699  < 2e-16 ***
data$Research                         -0.0104136  0.0167831  -0.620 0.535229
data$University.Rating                -0.0005529  0.0046889  -0.118 0.906183
data$CGPA                              0.1174724  0.0095195  12.340  < 2e-16 ***
data$TOEFL.Score                       0.0028743  0.0008646   3.325 0.000952 ***
data$LOR                               0.0175750  0.0039323   4.469 9.75e-06 ***
data$GRE.Score                         0.0017693  0.0005009   3.532 0.000451 ***
data$Research:data$University.Rating   0.0121379  0.0053915   2.251 0.024806 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.05969 on 492 degrees of freedom
Multiple R-squared:  0.8237,    Adjusted R-squared:  0.8212
F-statistic: 328.3 on 7 and 492 DF,  p-value: < 2.2e-16
```

- Checking validity assumptions for the Step Model




```
        Shapiro-Wilk normality test

data:  res1
W = 0.93043, p-value = 1.742e-14
```

B Bibliography

- Mohan S Acharya, Asfia Armaan, Aneeta S Antony :*A Comparison of Regression Models for Prediction of Graduate Admissions*, IEEE International Conference on Computational Intelligence in Data Science 2019,*Graduate Admission 2*;
- Marianne Jonker Fetsje Bijma and Aad van der Vaart. *An Introduction to Mathematical Statistics.* 2nd ed Epsilon Uitgaven, 2016;
- R Documentation;
- Zach Bobbit, *K-Fold Cross Validation in R (Step-by-Step)*;